BHANU PRASAD NELIPUDI

Aerospace Design Engineer | Airframe Structures | Aircraft Tooling | Design for Manufacturing
Bremen, Germany | bhanuprasadnelipudi@gmail.com | linkedin.com/in/bhanuprasadnelipudi | bhanuprasadnelipudi.github.io

PROFILE

Aircraft design engineer with 2+ years of professional experience across the aircraft lifecycle, from conceptual and detailed design to tooling, ground support equipment, structural inspection and maintenance. Experience on Airbus A330/A350 structural modifications, Dassault Rafale fuselage tooling, fighter-aircraft structural overhaul at HAL and a two-seater aircraft programme, plus an earlier land-vehicle suspension design study and a port-engineering internship. B.Tech in Mechanical Engineering with Mechanics of Solids, Design of Machine Elements and Finite Element Analysis; M.Eng. in Aeronautical Management (Hochschule Bremen) completing in March 2027. Goal: become an aircraft design engineer who can design, justify and deliver airframe structures for the next generation of sustainable aircraft. Open to a Master's thesis or internship in aircraft design and airframe structures.

CORE SKILLS

Design and CAD: CATIA V5 (Part Design, Assembly, Surface, Sheet Metal, Drafting); 3D modelling, 2D drafting, parametric design; GD&T; large assemblies of 1000+ parts; CAD quality checks and design reviews; Siemens NX (currently learning)

Structures and Analysis: Finite Element Analysis; NASTRAN/PATRAN (basic structural analysis); Mechanics of Solids; Strength of Materials Lab; Material Science and Metallurgy; load-path understanding from fighter-aircraft overhaul; MATLAB (basic)

Manufacturing and Maintenance: Design for Manufacturing (DfM); machining fundamentals; jigs, fixtures, walkway platforms, lifting equipment and ground support equipment; structural inspection, defect marking, snag reporting; repair and upgrade support; 5S

Standards and Tools: Airbus design standards (RSDP); Arina, Trend, Gilda; EASA CS-23 and SC-VTOL awareness; MS Office (Word, Excel, PowerPoint)

Working Style: team player; ownership of tasks; delivery under tight timelines; clear technical documentation

PROFESSIONAL EXPERIENCE

Design Engineer - Expleo Solutions Limited, Bangalore, India *May 2023 - Aug 2024*

Airbus Programs (A330 / A350)

- Delivered structural modification designs for the Airbus A330 and A350 using Arina, Trend and Gilda, in compliance with RSDP and Airbus design standards
- Produced 3D CATIA V5 models and 2D production drawings with parametric design integrity and clean model architecture; applied DfM and machining principles to improve manufacturability and first-time-right output

Dassault Aviation - Rafale Fuselage Tooling

- Converted 1000+ legacy 2D drawings into fully defined 3D CATIA models and updated the 2D documentation within a 2-month timeline
- Designed and validated jigs, fixtures and a walkway platform (including lifting equipment) for the Rafale fuselage; assembled 1000+ parts into complete tooling assemblies released for manufacturing
- Performed CAD quality checks and design reviews on colleagues' models before release; worked daily with an international team (India, France, UK) to clarify design intent and resolve production issues

Apprentice, Structural Overhaul - Hindustan Aeronautics Limited (HAL), Bangalore, India *Dec 2022 - May 2023*

- Studied fighter-aircraft structural layouts and load distribution through OEM manuals and overhaul documentation
- Performed structural inspections on in-service aircraft, identified and marked defective structural areas, and prepared snag reports for corrective action
- Supported and coordinated maintenance teams of 5-10 technicians during repair and upgrade activities, including material procurement and accurate drawing and manual compliance
- Supported a structural modification and upgrade programme and assisted engine ground-test engineers during full-power testing; applied 5S standards

Graduate Engineer Trainee - Bangalore Aircraft Industries Pvt. Ltd., Bangalore, India *Jun 2022 - Dec 2022*

- Completed accelerated structural design training: 2D-3D modelling, GD&T, detailed drafting and CATIA V5 (Part, Surface, Assembly, Sheet Metal)
- Supported the company's two-seater aircraft programme: structural design, part modelling and assembly validation; basic structural analysis in NASTRAN/PATRAN to support design feasibility
- Independently designed a wing support structure from concept to detailed design; designed ground support equipment components and an aircraft wing maintenance ladder
- Learned from senior scientists and engineers to design to constraints (budget, material, strength, fit, usability, manufacturability) instead of over-designing

ADDITIONAL INDUSTRY TRAINING

- Aircraft Design Internship (online), MARS Exploration Pvt Ltd (Mahamanav Aerospace and Research Space Exploration), Aug - Nov 2022
- Mechanical Internship, M&EE Department, Visakhapatnam Port Trust, May 2019: exposure to ships and the vehicles and equipment used in port operations
- CATIA V5 online training, Andhra Pradesh State Skill Development Corporation, June 2020
- Industrial Training, B.Tech curriculum, 2021

EDUCATION

M.Eng. in Aeronautical Management - Hochschule Bremen, Germany *Oct 2024 - Mar 2027 (expected)*

Full-time, taught in English. Completed modules (42 ECTS): International Law / Airlaw; Air Transport Business Administration; Management Systems (aircraft development process, project planning and execution, quality management systems); Human Factors in Leadership; Modelling and Simulation (flight physics, equations of motion, aircraft simulation model developed in a team project); Risk Management in Airline Operation; Simulation of Logistic Aviation Systems.

B.Tech. in Mechanical Engineering - Anil Neerukonda Institute of Technology and Sciences, India *Aug 2017 - Jan 2022*

Project Phase I and II (basis of the published research paper below).

Structures and Design: Mechanics of Solids; Engineering Mechanics; Strength of Materials Lab; Design of Machine Elements I and II; Theory of Machines I and II; Finite Element Analysis; Computer Aided Design and CAD Lab; Machine Drawing; Engineering Drawing; Advanced Engineering Drawing

Materials and Manufacturing: Material Science and Metallurgy; Manufacturing Technology I, II and III with Labs I and II; Metrology and Mechatronics Lab; Workshop; Mechanical Engineering Labs I and II

Thermo-Fluids and Systems: Engineering Thermodynamics I, II and III; Fluid Mechanics; Heat Transfer; Heat and Mass Transfer Lab; Hydraulic Machinery and Systems; Power Plant Engineering; Refrigeration and Air Conditioning; Instrumentation and Control Systems

Management: Industrial Engineering and Management; Operations Research; Managerial Economics and Financial Analysis

CERTIFICATIONS AND PUBLICATION

- NPTEL (IIT, India), proctored: Additive Manufacturing; Digital Manufacturing Technologies; Automation in Manufacturing; CAD/CAM; Robotics (Wheeled Mobile Robots)
- Publication: Design and Analysis of In-Wheel Suspension System, IJRASET, 2022 (structural design, modelling and analysis of land-vehicle suspension components)

LANGUAGES

English: C1 (fluent; Master's studies taught in English). German: A1, learning. International experience: relocated from India to Germany for full-time Master's study; worked daily with colleagues in France and the UK.